Lorraine Nien 粘清涵

Cyber-Physical Systems & Digital Twin Engineer

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Education

MS **National Cheng Kung University (NCKU)**

Master of Architecture (Digital) *2023-Present*

- **GPA: 3.82 / 4.3**
- **Coursework: Robotic Fabrication, Computational Design, Unity, Machine Learning, Architecture Design.**
- **Focus on Robotics Systems, Edge AI, and Cyber-Physical Systems for Built Environments**

BS **Tung Hai University (THU)**

Bachelor's Degree in Landscape Architecture *2019 - 2023*

Skills

AEC Domain Knowledge

- **BIM Data Integration:** Expert in extracting Revit data via APS (Autodesk Platform Services) for IoT application
- **Computational Design: Rhino & Grasshopper** for parametric modeling and spatial logic.
- **Physical Constraints:** Deep understanding of construction site layouts and MEP systems (crucial for robot navigation).

Robotic & Simulation

- **ROS2:**Concepts of Node communication, Topics, and Services for system architecture.
- **Unity 3D:** Developing Digital Twins and physics-based simulations for robot environments.
- **Foxglove/Rviz:** Real-time sensor data visualization and system debugging.

Soft Skills

- **Collaborative Problem Solving:** Experienced in coordinating with diverse stakeholders to resolve system integration conflicts.
- **Analytical Thinking:** Data-driven approach to troubleshooting and optimizing automated workflows.
- **Process Optimization:** Dedicated to improving operational efficiency through automation and SOP standardization.

Professional Experience

National Cheng Kung University (NCKU) SYNC Lab

Full-time Research Assistant *Nov 2025- Feb 2026*

Experience: APS Digital Twin Platform

- **Architected Digital Twin Solution:** Built a full-stack web application using React.js and Autodesk APS, transforming static BIM models into real-time 3D dashboards for facility management.
- **Integrated Multi-Source Data:** Engineered pipelines to visualize ESG metrics (Carbon/Energy) and AI-powered CCTV streams, enabling data-driven sustainability strategies.
- **Enhanced System Usability:** Designed a responsive UI to synthesize heterogeneous IoT data, reducing cognitive load and improving operational decision-making efficiency.

THESIS

SAFER-ROBIM: Autonomous Mobile Robot (AMR) for Industrial Safety *Master's Thesis 2025*

- **Built End-to-End AMR System:** Developed a custom autonomous mobile robot using **ROS 2 Humble** and **Sensor Fusion** (2D LiDAR + Depth Camera) for robust SLAM navigation in dynamic indoor environments.
- **Deployed Edge AI:** Fine-tuned a **YOLOv11** model and deployed it on **NVIDIA Jetson Orin Nano**, utilizing **TensorRT** optimization to achieve real-time hazard detection (e.g., Fire, Falls) with **91% precision**.
- **Bridged Physical & Digital:** Implemented a bi-directional **WebSocket** data bridge to sync robot telemetry with the BIM Digital Twin, automating inspection reporting and 3D hazard visualization.